Roman Merkulov

Python Backend & AI Engineer

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Technical skills

Languages: Python, TypeScript/JavaScript, SQL, Java; working knowledge of Rust, Bash, and PowerShell
Backend and APIs: FastAPI, Flask, REST, OpenAPI, Pydantic, asynchronous processing, service integrations
Cloud and infrastructure: AWS S3, SQS, CloudWatch, DocumentDB; Docker, Linux, Git
Databases: PostgreSQL, MongoDB, Amazon DocumentDB, Redis, MariaDB
AI and data: PyTorch, scikit-learn, NumPy, Pandas, spaCy, transformer models, LLM APIs, prompt design, document classification, structured extraction
Frontend: React, TypeScript, JavaScript, HTML/CSS
Security: Burp Suite, web/API security testing, OAuth and JWT, OWASP vulnerability classes, threat modeling, recon automation

Professional experience

Software Engineer — Apr. 2026–Present
FSM Rechtsanwälte — Vienna, Austria

- Develop AWS-based components connecting Python backend services to React/TypeScript frontends.
- Evaluate LLM-based retrieval and document-review workflows to speed up initial document screening.
- Conduct basic technical and security reviews of websites as part of legal workflow screening.
- Improve database queries and application workflows that support document processing and screening.

Lead Software Engineer — Mar. 2025–Mar. 2026
Schweinhammer Rechtsanwalt — Vienna, Austria

- Designed and delivered an AI-assisted document-processing application with Python backend services, React/TypeScript interfaces, MongoDB/DocumentDB, and AWS.
- Built asynchronous OCR, document-classification, and structured-extraction workflows using Amazon SQS and commercial and open-source LLMs.
- Developed APIs, validation rules, state-management workflows, and interfaces for legal and procurement documents.
- Integrated external verification services and handled successful, expired, missing, and failed verification states explicitly.
- Owned technical architecture and implementation across backend, frontend, AI, and infrastructure components.

Freelance Software Engineer — May 2023–Jan. 2025
Fintech client project — Remote

- Developed Python web services and REST APIs for fintech and banking-related workflows.
- Built AWS-hosted application components and integrations between frontend applications, backend services, databases, and external providers.
- Integrated LLM-based functionality for document processing, user assistance, and application automation.
- Improved error handling, testing, and maintainability across cloud-hosted services.
- Supported deployment, service configuration, and production debugging of distributed application components.

Earlier experience

Machine Learning Intern Jan. 2022–Nov. 2022
NLABTEAM Remote

- Developed and improved NLP pipelines for chatbot applications, including text preprocessing, intent classification, entity extraction, and response workflows.
- Used spaCy and transformer-based models to improve natural-language-understanding components.
- Evaluated chatbot output and diagnosed issues in classification, retrieval, and response relevance.
- Tested model configurations and preprocessing strategies, then integrated ML components into the existing chatbot architecture.

Technical Consultant Apr. 2020–Jun. 2022
Independent Vienna / Remote

- Developed and optimized Python ETL pipelines for client data workflows.
- Supported migration from monolithic applications to Docker- and Kubernetes-based service architectures.
- Advised on backend architecture, infrastructure security, deployment practices, and secure data handling.

Selected technical projects and research

Distributed Data-Processing Platform | Rust, Python Jul. 2026–Present

- Developing a backend service for uploading and querying CSV and telemetry datasets through an HTTP API.
- Using Rust, Axum, Tokio, and Serde for asynchronous request handling, data validation, and structured JSON responses.
- Using Python to generate test datasets, validate API results, and build integration and performance tests.
- Extending the architecture with analytical query execution, background job processing, persistent metadata storage, and observable failure handling.

Autonomous AI-agent security research

- Research autonomous-agent security boundaries, including prompt injection, persistent-memory poisoning, unsafe tool invocation, and authorization failures.
- Built an isolated ARM-based VM and Docker lab for vulnerability reproduction and defensive monitoring.
- Applied STRIDE threat modeling to agent components, tools, memory, communication channels, and trust boundaries.
- Reproduced published vulnerabilities and documented allow/deny policies, telemetry requirements, and security controls.

Education

B.Sc. Computer Science, in progress 2022–2026
University of Vienna Vienna, Austria
Bachelor thesis: Security of Autonomous AI Agents: prompt injection, memory poisoning, and tool authorization.

B.Sc. Physics Sept. 2018–June 2022
Moscow Institute of Physics and Technology (MIPT) Moscow, Russia

Languages

Russian (native) | English (C1) | German (C1)